
I Introduction

Snow is one of the major component of the cryosphere. Its global distribution on both ocean and land areas strongly depends on seasonality, latitude, and elevation. It occurs in three forms, mainly: permanent snow on ice sheets and glaciers, snow on sea ice, and seasonal snow cover. The latter is estimated to account for 1.3% (summer) to 30.6% (winter) of the Northern Hemisphere land surface (Vaughan et al., 2013), with the 1967-2026 mean snow cover extent (SCE hereafter) in winter being around $39.79 \times 10^6 \text{ km}^2$ (Robinson & Estilow, 2012). Sea ice is almost completely covered by snow during the winter season; its peak concentration, therefore representative of sea ice specific SCE, occurs in March in the Northern Hemisphere ($15.03 \times 10^6 \text{ km}^2$) and in September in the Southern Hemisphere ($18.61 \times 10^6 \text{ km}^2$) (Fetterer et al., 2025; Warren et al., 1999). Finally, perennial snow mean extent over the 2000-2022 period is estimated at $16.47 \times 10^6 \text{ km}^2$ from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS), including glaciers, ice caps, Greenland and Antarctica ice sheets (Johnston et al., 2024).

These several tens of millions of snow-covered squared kilometres play a major role in freshwater resources availability, ecosystems equilibrium and, particularly, climate regulation. In fact, among its many physicochemical properties, snow represents one of the strongest reflective materials on Earth's surface. The fraction of solar radiation delivered to a body that is diffusely reflected by it is called *albedo*; for snow, this amount is roughly close to 1, since its highest reflectivity occurs in the range of visible light, *i.e.* where solar radiation intensity peaks (Kokhanovsky & Zege, 2004). Thanks to its high albedo, snow represents of the main contributors to Earth's energy budget regulators (National Snow and Ice Data Center, 2024). Of the 340 W m^{-2} of solar radiation incident at the top of the atmosphere, Northern Hemisphere land snow cover alone reflects approximately 2 W m^{-2} back to space, averaged over the 1979–2008 period; in May, when SCE and soil insulation are highest, this quantity can reach 4.5 W m^{-2} (Flanner et al., 2011).

As the whole cryosphere in general, snow distribution on Earth is undergoing dramatic changes because of the current anthropogenic perturbation of the climate system and warming of the atmosphere. At the global scale, SCE is experiencing a negative trend since the 1980s, with a statistically significant reduction in the mean area of 5% after 1988, compared to the 1967-1988 period, and an earlier melt onset in the spring season, in the Northern Hemisphere only (Dye, 2002; Robinson & Frei, 2000). Less snow cover in both space and time, driven by rising temperatures, reduces terrestrial albedo and increases the absorption of incoming solar radiation at the surface, which in

turn amplifies warming (Cubasch, U., 2001; Holland, M. M. & Bitz, C. M., 2003). This mechanism is known as the positive snow-albedo feedback: it is estimated to be responsible of a reduction in Northern Hemisphere land snow cover radiative cooling forcing of $\sim 0.22 \text{ W m}^{-2}$ from 1979 to 2007, which nearly the same contribution of Arctic sea ice (Flanner et al., 2011).

its intensity is amplified even more when light-absorbing particles, such as dust, black carbon and microbial growth, .

II Material and methods

1 Theoretical backgrounds

!!! This risks to become "textbook"-like - then, must be SUPER ADAPTED TO MY SPECIFIC GOAL AND BE VERY PERSONAL AND NOT GENERAL AT ALL - PASSIONATING ! write it after matmethods and results

2 Material and instruments

2.1 *In situ*: the Multiband albedometer

2.1.a *The stations*

Cap Prud'Homme

EAIIST

2.2 *In situ*: the Solalb albedometer

2.2.a *Campaigns*

2.3 Satellite imagery: Sentinel-3 Ocean Land and Colour Instrument

2.3.a *Generalities*

Orbit, resolution, swath, revisit time etc.

2.3.b *Data availability*

3 The workflow

Nice flowchart

3.0.a

III Results

PRESENTING MY RESULTS CSI Problem horizontality/titling ? Calibrations analysis/response symmetry

Do not simply list the results, it can be boring If the conclusion from a result is very fast, just put it in the results

IV Discussion

CONTEXTING MY RESULTS IN SCIENTIFIC LITERATURE Typically no plots

V Conclusion

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References

- Cubasch, U. (2001). Projections of future climate change.
- Dye, D. G. (2002). Variability and trends in the annual snow-cover cycle in Northern Hemisphere land areas, 1972–2000. *Hydrological Processes*, 16(15), 3065–3077. <https://doi.org/10.1002/hyp.1089>
- Fetterer, F., Knowles, K., Meier, W. N., Savoie, M., Windnagel, A., & Stafford, T. (2025). *Sea Ice Index, Version 4*. National Snow and Ice Data Center. <https://doi.org/10.7265/A98X-0F50>
- Flanner, M. G., Shell, K. M., Barlage, M., Perovich, D. K., & Tschudi, M. A. (2011). Radiative forcing and albedo feedback from the Northern Hemisphere cryosphere between 1979 and 2008. *Nature Geoscience*, 4(3), 151–155. <https://doi.org/10.1038/ngeo1062>
- Holland, M. M. & Bitz, C. M. (2003). Polar amplification of climate change in coupled models. 21, 221–232.
- Johnston, J., Jacobs, J. M., & Cho, E. (2024). Global Snow Seasonality Regimes from Satellite Records of Snow Cover. *Journal of Hydrometeorology*, 25(1), 65–88. <https://doi.org/10.1175/JHM-D-23-0047.1>
- Kokhanovsky, A. A., & Zege, E. P. (2004). Scattering optics of snow. *Applied Optics*, 43(7), 1589–1602. <https://doi.org/10.1364/AO.43.001589>
- National Snow and Ice Data Center. (2024). What is the cryosphere? <https://nsidc.org/learn/what-cryosphere>
- Robinson, D. A., & Estilow, T. W. (2012). Northern hemisphere snow cover extent dataset. NOAA National Centers for Environmental Information. <https://climate.rutgers.edu/snowcover/>
- Robinson, D. A., & Frei, A. (2000). Seasonal Variability of Northern Hemisphere Snow Extent Using Visible Satellite Data. *The Professional Geographer*, 52(2), 307–315. <https://doi.org/10.1111/0033-0124.00226>
- Vaughan, D., Comiso, J., Allison, I., Carrasco, J., Kaser, G., Kwok, R., Mote, P., Murray, T., Paul, F., Ren, J., Rignot, E., Solomina, O., Steffen, K., & Zhang, T. (2013). Observations: Cryosphere. In T. Stocker, D. Qin, G.-K. Plattner, M. Tignor, S. Allen, J. Boschung, A. Nauels, Y. Xia, V. Bex, & P. Midgley (Eds.), *Climate change 2013: The physical science basis. Contribution of working group I to the fifth assessment report of the intergovernmental panel on climate change*. Cambridge University Press.
- Warren, S. G., Rigor, I. G., Untersteiner, N., Radionov, V. F., Bryazgin, N. N., Aleksandrov, Y. I., & Colony, R. (1999). Snow Depth on Arctic Sea Ice. *Journal of Climate*, 12(6), 1814–1829. [https://doi.org/10.1175/1520-0442\(1999\)012<1814:SDOASI>2.0.CO;2](https://doi.org/10.1175/1520-0442(1999)012<1814:SDOASI>2.0.CO;2)